Advanced functions

NULL functions

- no value - no data - no records
- missing value - empty space

• these functions help us to deal with Null value in our data set.

Case 1: how to check for NULL values in a specific table or a specific column.

Fruit sales

SaleID	Product	Amount
1	Banana	100
2	Apple	NULL
3	Orange	20

IS NULL function

Syntax : checks if you have a NULL value on column

```
SELECT    columns)
FROM      table_name
WHERE     column_name is NULL;
```

Case 2: we want to find the rows that does not have NULL values.

IS NOT NULL function (filters the rows)

Syntax :
```
SELECT *
FROM table_name
WHERE column_name is NOT NULL;
```

~~Case 3: replacing a NULL value in a specific column in a spec~~

Case 3: Replacing a NULL value with a default value.

IF NULL (column_name, replacement_value)

Place within a SELECT statement used with AS to give it a descriptive name.

Syntax:

```
SELECT column(s),
IF (amount, 0) AS amount
FROM table_name;
```

This will replace the NULL with 0

Case 4: we want to have multiple alternative values

Coalesce - advises if theres is a null, put another ~~ans~~

- returns the first Non-null value from a list of expressions
- it checks the expression from left-right.

Syntax

```
SELECT column A
COALSE (column, colomn, 0) AS unique_id
FROM table.
```

boring_team

Name	student_id	ID_no
Rhoh	101	NULL
Mark	Null	111
Elon	103	112
John	102	NULL
Larry	NULL	114
Warren	NULL	NULL

SELECT name
coalesce (student_id, ID_no, 0) AS unique
FROM boring_team;

Name	Unique_ID
Rhoh	101
Elon	103
John	102
Larry	114
Warren	0

Date functions - found within the select statement.

- functions that enable to deal with dates
- indicates a specific period.
- indicates an event - creates a new column
- we need to have this function to understand when a transaction took place
- we have to us AS to rename the column

① YEAR() - it is used to extract the year on the date column

syntax SELECT YEAR('2025-06-02') AS year of birth

② Month() - extracts the month

③ DAY() - extracts the day.

④ DAY NAME() - extracts monday - Sunday based on the system calander (time zone)

④ Month name() - extracts month name from JAN - DEC based on system time zone (DMS)

⑤ DATEADD (unit, value, date) - it is used to add or substract a date interval

⑥ DATEDIFF (unit, date, date2) - used to calculate the difference between the 2 dates.
(date 2 - date 1 = value)

⑦ CURRENT_DATE() - gives the todays date based on the DMS setting.